

CycloProp: Advanced UAV Propulsion Challenge

Stage 1 - Preliminary Engineering Design Report

Kalash CR-1, a single cyclo rotor propulsion module. Team Kalash, VPKBIET. Team ID TM-5A7C41AF909.

Team and Submission Information

Particular	Details
Team ID	TM-5A7C41AF909
Team Name	Kalash
Institution / Organization	VPKBIET (category: UG)
Team Leader	Kartik Shirode
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Proposed Design Name	Kalash CR-1
Date of Submission	27 September 2026

1. Executive Summary

1.1 Proposed Cyclo rotor Concept

Design name:

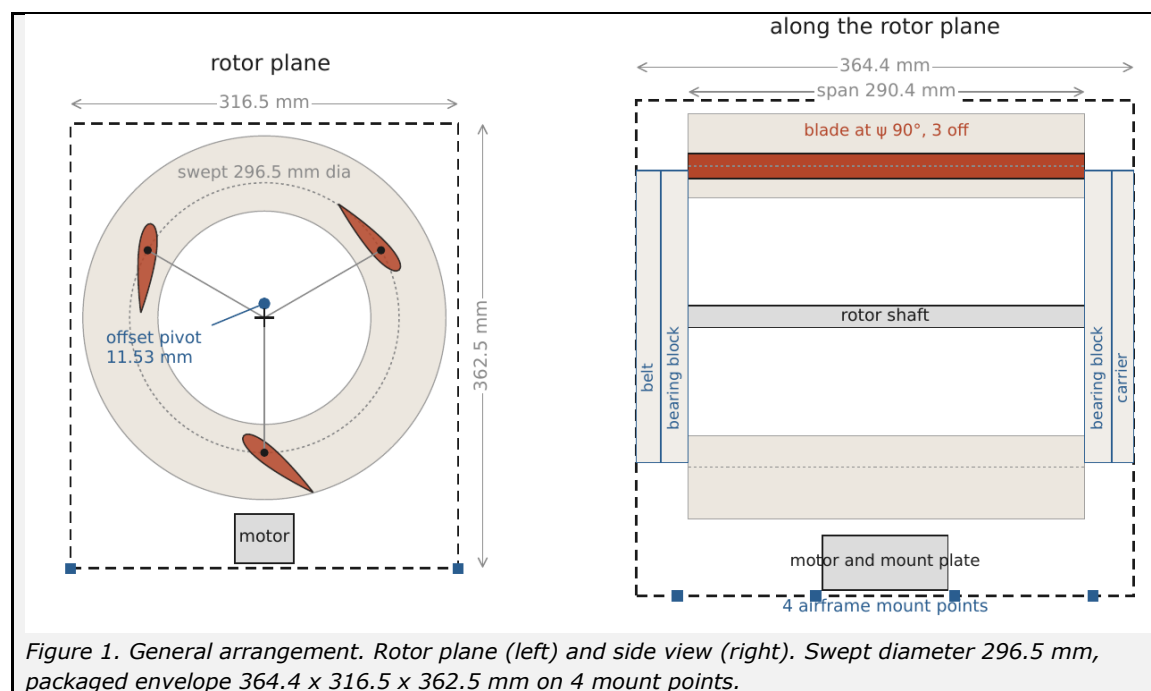
Kalash CR-1

Brief description of the proposed configuration:

One cycloidal rotor, 220.0 mm in diameter and 290.4 mm wide, with 3 NACA 0020 blades of 72.6 mm chord. Each blade pitches plus or minus 40 degrees about its 30 percent chord point through its own passive four-bar linkage, and all three linkages share one offset pivot 11.53 mm from the rotor axis.

Turning that offset about the rotor axis turns the thrust vector. Two 20 g servos do it through a phasing carrier outside the rotor, so nothing on the rotor is actuated. A T-Motor MN5006 KV450 drives the rotor at 2337 rpm through a 4.25 to 1 toothed belt from an 8S pack. Design thrust is

17.0 N.



1.2 Key Design Parameters

Parameter	Preliminary Value	Unit
Rotor diameter	220.0 (296.5 swept by the pitching blades)	mm
Rotor width	290.4	mm
Number of blades	3	--
Blade chord	72.6	mm
Blade profile / airfoil	NACA 0020	--
Operating RPM	2337	rpm
Blade pitch range	-40 to +40	deg
Pitch amplitude	40	deg
Estimated thrust	17.0 (16.15 on the low coefficient)	N
Estimated shaft power	377.4	W
Estimated electrical power	518.0 at the module	W
Estimated module mass	0.688 (0.776 conservative)	kg
Estimated thrust-to-weight ratio	2.5191 (2.1221 worst case)	--
Estimated maximum thrust-vectoring angle	+/- 60 (120 of authority)	deg

1.3 Design Summary

Principal design features:

One rotor rather than a cluster. On a common model it gives the lowest module mass and the best thrust to weight of the three layouts we compared (section 3).

Passive cyclic pitch. A four-bar per blade sets the 40 degree amplitude from one 11.53 mm offset, and no actuator rides on the rotor.

Thrust vectoring by rotating that offset: 120 degrees of authority from 2 servos through a 1.5 step up gear pair.

Sandwich blades of PMI foam, 2 plies of 60 gsm carbon twill and a CFRP spar on the pitch axis, 31.75 g each.

Every number in this report comes out of Python scripts reading one data file, and a checking script recomputes them, so geometry, performance and mass stay consistent.

Key design innovation / distinguishing feature:

The pitch schedule and the thrust direction come from the same linkage. The offset length sets the amplitude and its direction sets the phase, so vectoring needs one small rotation of a carrier that doesn't spin with the rotor.

The linkage solution also decided the drivetrain. The solved pitch links pass within 0.019 mm of the rotor axis, so no shaft can run through that plane, and the rotor is driven from one end only.

Expected performance:

17.0 N at 2337 rpm for 518.0 W at the module. The mass budget is 687.91 g, so thrust to weight is 2.5191 at the design point. With the conservative mass and a 5 percent thrust loss for blade flexibility applied together it falls to 2.1221.

Principal technical challenges identified:

1. Thrust to weight clears 2.5 by only 0.8 percent, on a mass estimate that isn't accurate to 0.8 percent. Closing the worst case needs 117.26 g.
2. The thrust coefficient is transferred from published rotors of the same shape family, not measured on this one.
3. The motor runs at 93 percent of a derated continuous power rating.
4. The quasi steady model almost certainly under-predicts side force, which costs vectoring authority.

2. Requirements and Design Targets

2.1 Challenge Requirements

Parameter	Challenge Requirement	Team Design Target
Thrust	≥ 10 N	17.0 N design, 16.1493 N with a 5% blade flexibility loss
Thrust-to-weight ratio	> 2.5	2.5191 design. 2.1221 with both downsides. Stage 2 gate: conservative mass under 658.48 g
Rotor type	Cycloidal rotor with blade-pitch variation	Single cyclorotor, 3 blades, passive four-bar cyclic pitch, ± 40 deg
Thrust vectoring	Required	120 deg of direction authority by rotating the pitch offset
Design approach	CAD and engineering analysis	Closed form analysis in Python from one data file. CAD, CFD and FEA are Stage 2 items
Final objective	Build-ready design for subsequent implementation	Geometry frozen at Stage 1. Stage 2 turns it into CAD, drawings, a verified mass and a spin test plan

2.2 Selected Design Targets and Rationale

Design Parameter	Selected Value	Basis / Engineering Rationale
Rotor diameter	220.0 mm	Smallest radius the selected motor holds continuously. At 100 mm the motor would need 531.5 W against 520.0 W derated continuous
Rotor width	290.4 mm	4 chords, blade aspect ratio 4, from Kellen's UAV scale optimum shape family [1]
Number of blades	3	At fixed solidity fewer blades give more thrust [2], and 3 is Kellen's optimum for this Reynolds band [1]
Blade chord	72.6 mm	Chord to radius 0.66, Kellen's optimum. Solidity 0.3151, inside the 0.30 to 0.40 band he measured
Blade profile / airfoil	NACA 0020	Thick symmetric section. Efficient at this Reynolds number and deep enough for a real spar
Operating RPM	2337	The speed at which a blade area coefficient of 0.6055 gives 17.0 N. Chord Reynolds 130,296, inside Kellen's 100,000 to 300,000

Mean blade pitch	0 deg	The horn to chord angle, -102.50 deg, is chosen so the cycle mean pitch is zero
Pitch amplitude	40 deg	Kellen's optimum for this shape family [1]
Maximum vectoring angle	+/-60 deg	80 deg of servo travel through a 1.5 step up gives 120 deg of carrier rotation
Target power	518.0 W	Motor input 483.2 W against the MN5006's 520 W derated continuous rating
Target module mass	under 693 g	The mass at which 17.0 N gives exactly 2.5. Budget 687.91 g nominal, 775.74 g conservative

2.3 Design Constraints and Assumptions

List only assumptions that directly affect geometry, thrust prediction, power estimate, mass estimate, structural assessment, or manufacturability.

No.	Constraint / Assumption	Basis
1	Blade area thrust coefficient 0.6055, held below the 0.6648 Kellen measured on this shape family	[1], [2]. Deliberately low
2	Figure of merit 0.5215: Kellen's 0.6 scaled to our lower coefficient by $(0.6055 / 0.6648)^{1.5}$	[1]
3	Efficiencies: belt 0.93, motor 0.84, ESC 0.95. Rotor tare 10 percent of shaft power	Efficiencies assumed; tare from [2]
4	Motor 180 s ratings derated to 0.80 for continuous duty	Judgement. The problem statement sets no endurance to size it against
5	Hover at sea level, ISA air, density 1.225 kg/m ³ , no freestream	Standard
6	Blade loads at 4.0 times the mean, a 1.20 overspeed case, every margin floored at 1.5	Top of the published 3 to 4 range [8]
7	Material allowables are published class values	No coupon tests yet
8	Mass growth by line class: 8% catalogue, 12% machined, 15% calculated, 25% allowance	Set before the total was known

3. Concept Development and Selection

Teams may present up to three concepts. If only one concept was considered, provide the design rationale and trade-offs clearly.

3.1 Concept Options

Item	Concept 1	Concept 2	Concept 3
Concept name	Single rotor, 3 blades, R 110 mm	Two rotors, 3 blades each, R 80 mm	Three rotors, 3 blades each, R 65 mm
Operating principle	One rotor makes all 17.0 N at 2337 rpm	Each rotor makes 8.5 N at 3124 rpm	Each rotor makes 5.67 N at 3864 rpm
Advantages	Least hardware: one shaft, one mechanism, one belt. Chord Reynolds 130,296, inside the measured band. Lowest mass	Smaller rotors. Aerodynamic power 330.3 W, 3% lower. Differential thrust possible	Smallest rotors. Aerodynamic power 331.93 W. Most control options
Limitations	Largest single rotor and the most torque, 1.5424 Nm, so it needs a belt reduction	Hardware doubles. Reynolds 92,133, below the measured band. 400 mm largest dimension	Hardware triples. Reynolds 75,227. 490 mm largest dimension, three mechanisms to phase together
Expected performance	647.23 g, T/W 2.6774, stacked downside 2.1281	796.12 g, T/W 2.1767, stacked downside 1.7286	994.94 g, T/W 1.7417, stacked downside 1.3823

End views to a common scale, 17.0 N total thrust in each

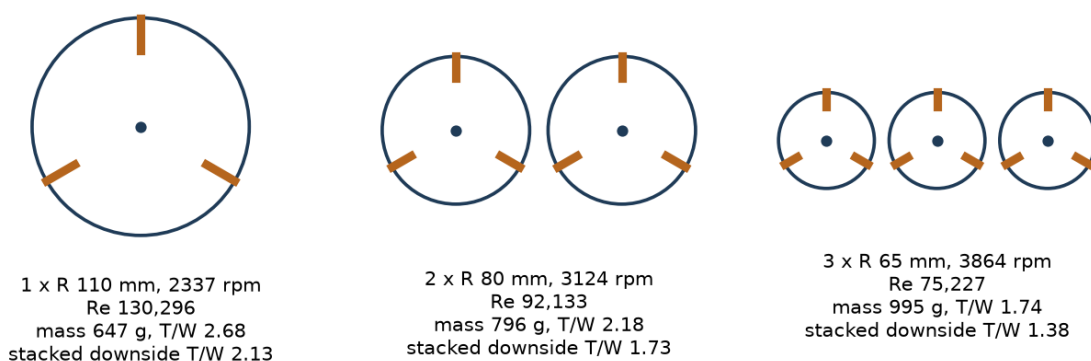


Figure 2. The three layouts, end views to one scale. Masses are the early mass envelope applied the same way to all three; the refined budget in section 8 applies to the selected concept only.

3.2 Concept Selection Matrix

Evaluation Parameter	Weight (%)	Concept 1	Concept 2	Concept 3
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Thrust potential	15	5	3	2
Power requirement	10	4	5	5
Mass	20	5	3	1
Pitch-control capability	10	5	3	2
Thrust-vectoring capability	10	4	4	4
Structural feasibility	10	4	4	4
Manufacturability	10	5	3	2
Mechanism complexity	10	5	3	2
Cost	5	5	3	2
Total	100	94	68	50

Selected concept:

Concept 1, the single rotor.

Basis for selection:

It wins the criterion the challenge makes hard, module thrust to weight, by 23 percent on the stacked downside case (2.1281 against 1.7286), and it wins every other column apart from aerodynamic power, where the clusters are about 3 percent better. Splitting the thrust splits the aerodynamics but not the hardware: 7 of the 13 mass envelope lines multiply by rotor count. Every disputed assumption was set in the clusters' favour, with no wake interaction, one shared motor and controller, and the same coefficient despite their lower Reynolds number.

Scores are 1 to 5 and the total is the weighted sum out of 100: 94, 68 and 50. The thrust, power and mass rows follow the numbers in 3.1. The other rows are our judgement and are marked as such here.

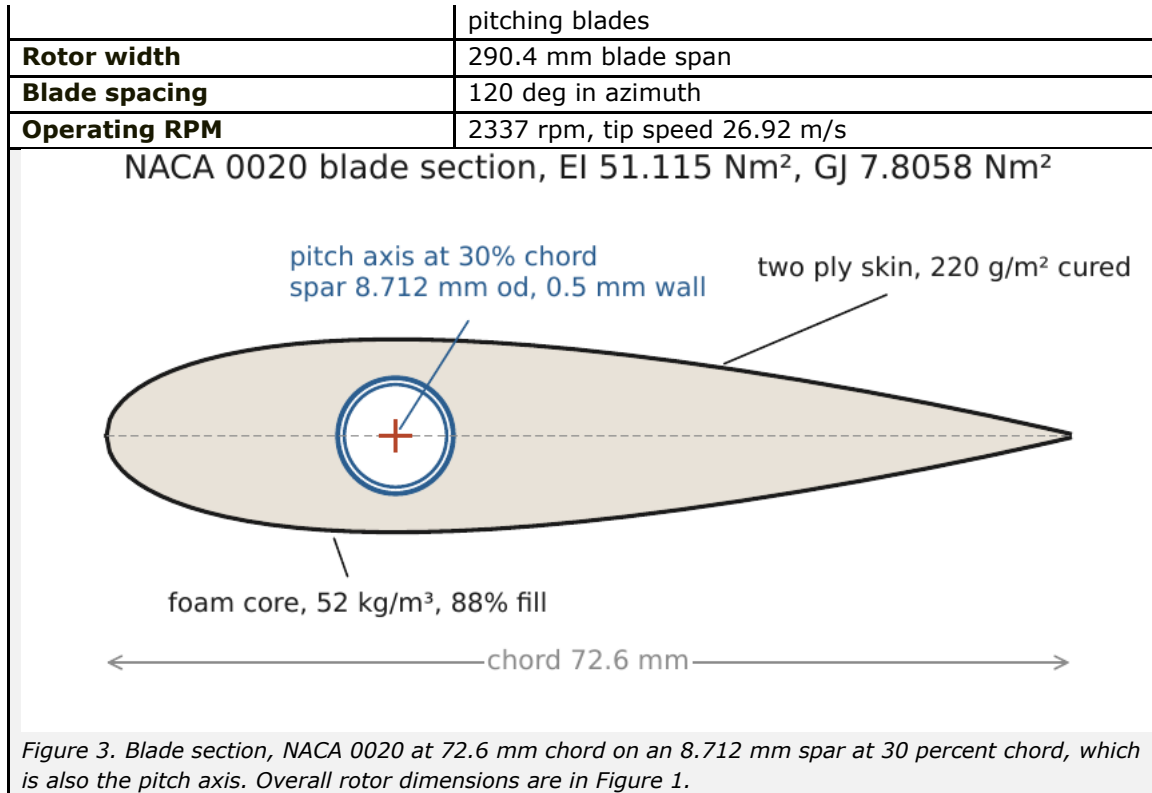
Key trade-offs:

A single rotor needs 1.542 Nm at 2337 rpm, more than any outrunner in this mass class gives directly, so it carries a belt reduction. It also has the largest swept diameter, 296.5 mm, which sets the envelope. We accept both for about 149 g less module mass than the twin.

4. Rotor and Blade Design

4.1 Rotor Configuration

Parameter	Description / Value
Rotor architecture	Single cyclorotor. Blades parallel to the rotation axis, pinned at both ends to CFRP spider arms on 7075-T6 hub bosses, on a CFRP shaft driven from one end
Number of blades	3
Direction of rotation	Counter-clockwise, looking along the axis from the non-drive end
Rotor diameter	220.0 mm to the pitch axes. 296.5 mm swept by the



4.2 Blade Geometry

Parameter	Value	Unit	Basis
Airfoil / blade profile	NACA 0020	--	Kellen's shape family [1]
Chord	72.6	mm	Chord to radius 0.66
Thickness	14.52	mm	20% of chord
Blade length	290.4	mm	Aspect ratio 4
Pitch range	-40 to +40	deg	Solved schedule reaches +40.52 and -39.35
Pitch amplitude	40	deg	Set by the 11.53 mm offset
Blade twist, if any	0 built	deg	0.014 under aerodynamic load; 2.2346 wind up at the free end from the centrifugal pitching moment
Estimated blade mass	31.75	g	Integrated from the section: core, skins, spar, 2 root fittings

Reason for blade-profile selection:

Thick symmetric sections hold their performance at chord Reynolds numbers near 130,000, and 20 percent thickness gives a 14.5 mm deep section: room for an 8.71 mm spar on the pitch axis inside a closed foam and skin cell. It is symmetric because the blade pitches both ways through 40 degrees. The curved flow at chord to radius 0.66 gives it virtual camber, which is already

inside the coefficient we transfer, since Kellen measured it on the same shape family.

Expected operating Reynolds-number range:

130,296 at the design point, from 1.225 kg/m³, 26.92 m/s and 72.6 mm chord. Over the operating cases in 5.4 it runs from about 113,942 to 130,296, inside Kellen's measured 100,000 to 300,000.

5. Preliminary Aerodynamic Analysis

5.1 Analysis Approach

Select the method used and provide sufficient details for the committee to reproduce or verify the estimate.

- ☒ Analytical calculation
- ☒ Blade Element / Blade Element Momentum Method
- ☒ Published experimental correlations
- ☐ CFD or other numerical method
- ☐ Other: _____

Item	Details
Software / calculation tool	Python 3 with NumPy and Matplotlib. tools/linkage.py (kinematics and loads), tools/structure.py (structure and mass), tools/check.py (recomputes and checks every number). All inputs and results in numbers.json
Operating condition	Hover, 17.0 N at 2337 rpm, no freestream
Air density	1.225 kg/m ³
Atmospheric conditions	ISA sea level. Viscosity 1.838e-05 Pa s, as implied by the stored Reynolds number
Other assumptions	Uniform induced inflow, quasi steady blade forces, no wake interaction, no dynamic stall beyond an incidence cap

5.2 Aerodynamic Calculation Summary

Calculation methodology:

Thrust comes from a blade area thrust coefficient measured on this shape family, not from a blade element model, because at chord to radius 0.66 a blade element model without a curvilinear flow correction is less trustworthy than the measurement. Power comes from momentum theory and a figure of merit, checked by a second route through a published power

loading. Blade loads come from a 36 point quasi steady azimuthal model run on the solved pitch schedule.

Governing equations used:

$T = C_{Tb} \times 0.5 \rho u^2 \times (N c b)$, with $u = \Omega R$ and $N c b$ the blade area

$P_{ideal} = T^{1.5} / \sqrt{2 \rho A}$, with $A = 2 R b$ the projected area; induced velocity $v_i = \sqrt{T / (2 \rho A)}$, inflow ratio $\lambda = v_i / u$

$FM = 0.6 \times (C_{Tb} / 0.6648)^{1.5}$; $P_{aero} = P_{ideal} / FM$; $P_{shaft} = P_{aero} / 0.9$ (10% tare)

$P_{elec} = P_{shaft} / (\text{eff}_{belt} \times \text{eff}_{motor} \times \text{eff}_{ESC})$; $Q = P_{shaft} / \Omega$; $Re = \rho u c / \mu$

Azimuthal load: $F(\psi) = 2 \pi \alpha \times (u_t^2 + u_p^2)$, $\alpha = \theta(\psi) - \text{atan}(u_p / u_t)$, α capped at 28 deg

Blade loading assumptions:

Mean vertical force 5.6667 N per blade. The azimuthal model gives a peak of 14.4260 N, a peak to mean of 2.5458. A quasi steady model under-predicts the peak, so the structure is sized on 4.0, the top of the published 3 to 4 range [8]. The blade is not chordwise balanced: its centre of mass sits at 39.31% chord against a 30% pitch axis, and the pitch loads carry that.

Airfoil data / aerodynamic coefficients used:

Blade area thrust coefficient 0.6055 nominal and 0.5752 low (5% flexibility loss). It is bracketed by 0.6648 measured by Kellen on this shape family [1], 0.7211 recomputed from Benedict's quad rotor hover point [2] and 0.8114 from his twin rotor, which we treat as an upside only. The load model uses a thin airfoil lift slope of 2π per radian, scaled to the measured mean.

5.3 Preliminary Performance Estimate

Parameter	Value	Unit
Tip speed	26.92	m/s
Reynolds number	130,296	--
Swept area	0.063888 (projected, 2R x b)	m ²
Estimated blade loading	0.6055 (blade area thrust coefficient)	--
Estimated thrust	17.0	N
Estimated torque	1.542	N m
Estimated shaft power	377.4	W
Estimated electrical power	518.0	W

Thrust estimation methodology and calculation:

$T = 0.6055 \times 0.5 \times 1.225 \times 26.92^2 \times 0.06325 = 17.0$ N, where 0.06325 m² is 3 blades of 72.6 by 290.4 mm. The low case takes 5 percent off for blade flexibility, 16.1493 N. Both clear 10 N on their own.

Power estimation methodology and calculation:

Ideal power over the projected area of 0.063888 m² is $17.0^{1.5} / \sqrt{2 \times 1.225 \times 0.063888} = 177.2$ W, with induced velocity 10.42 m/s and inflow ratio 0.3871. Over a figure of merit of 0.5215 that is 339.7 W at the blades. A 10 percent rotor tare, 37.7 W, gives 377.4 W at the shaft, 1.542 Nm at 2337 rpm. Through the belt, motor and ESC that is 508.6 W at the ESC, and the servos, controller and regulator loss take the module to 518.0 W. A second route through Benedict's measured power loading of 0.062 N/W gives 274.2 W at the blades, 19.3 percent lower. We carry the higher one.

Expected aerodynamic efficiency / figure of merit, if applicable:

Figure of merit 0.5215, Kellen's measured 0.6 scaled down to our lower coefficient so the same caution isn't spent twice. Power loading at the module is 0.0328 N/W.

5.4 Sensitivity to Operating Conditions

Provide estimates for at least three operating conditions.

Operating Condition	RPM	Pitch	Estimated Thrust	Estimated Power
Case 1: 13 N case	2044	+/-40 deg, mean 0	13.0 N	323.1 W motor input
Case 2: 16 N case	2267	+/-40 deg, mean 0	16.0 N	441.2 W motor input
Case 3: 17 N case (design point)	2337	+/-40 deg, mean 0	17.0 N	483.2 W motor input
Case 4: 18 N case	2405	+/-40 deg, mean 0	18.0 N	526.5 W motor input; over the 520 W rating

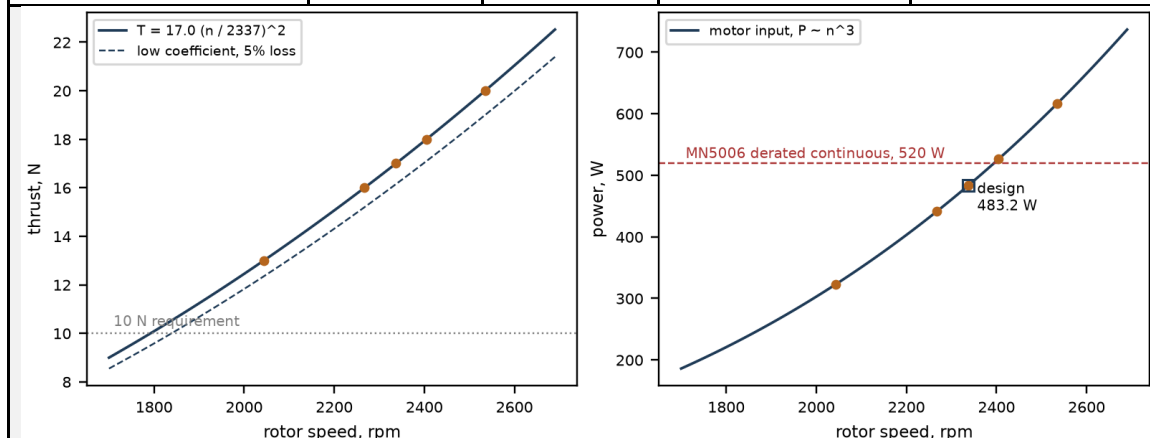
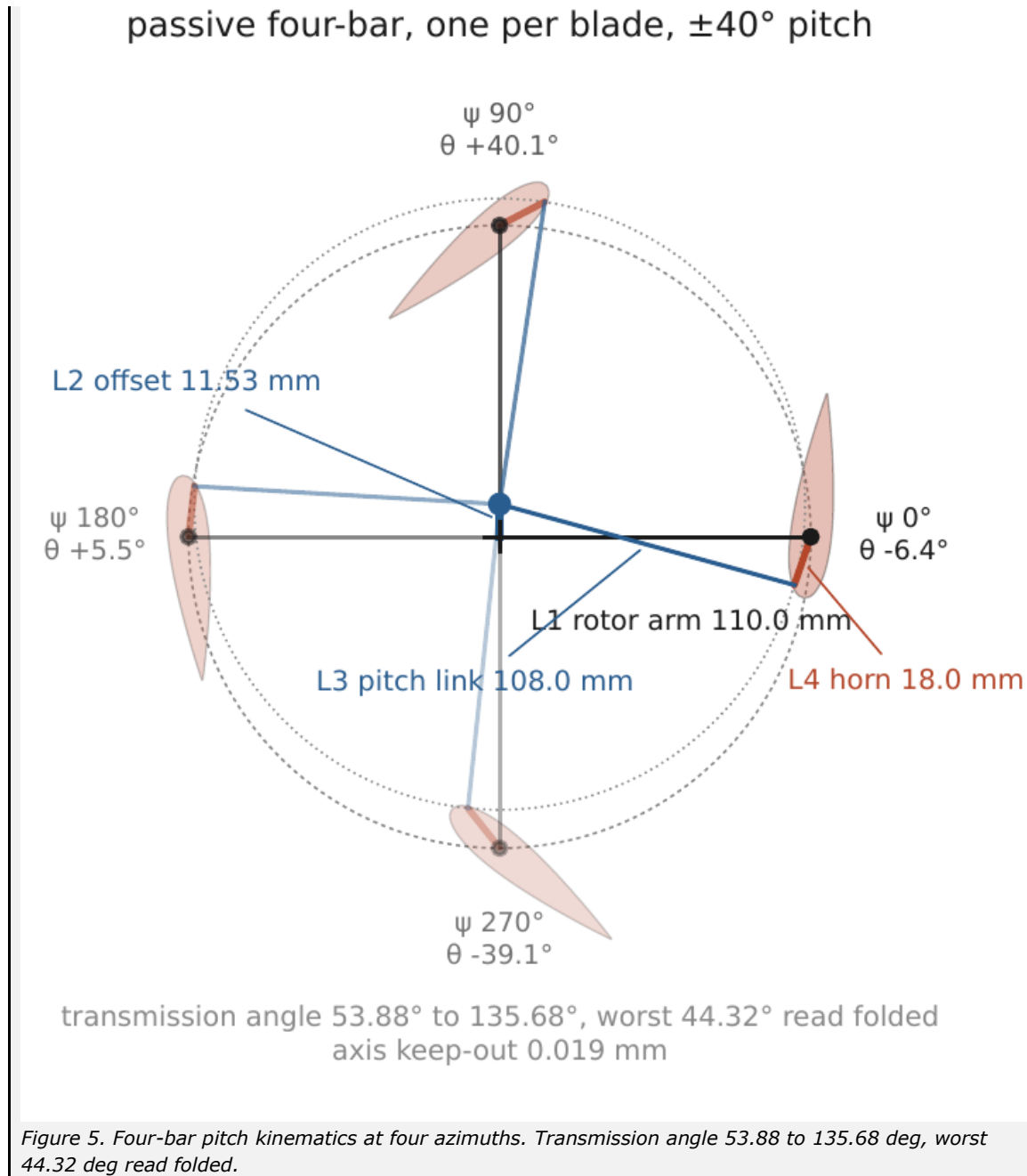


Figure 4. Thrust and motor input power against rotor speed at fixed pitch schedule. Points are the stored sensitivity rows (13, 16, 17, 18 and 20 N). Above the design point the motor runs out of continuous power, which is why the design stops at 17.0 N.

6. Blade-Pitch Kinematics and Thrust Vectoring

6.1 Pitch-Control Mechanism

Item	Description / Value
Mechanism description	Passive four-bar per blade: rotor arm L1 110.0 mm, offset link L2 11.53 mm (ground), pitch link L3 108.0 mm, horn L4 18.0 mm. All three blades share one offset pivot. Double crank by Grashof ($11.53 + 110.0 < 18.0 + 108.0$)
Actuation method	Rotor rotation drives the pitch. The actuators only rotate the offset direction, through a phasing carrier that doesn't spin
Actuator type	Digital metal gear servos, 20 g class, 3.9 kgf cm stall, on 60 mm sector gears driving a 40 mm carrier ring (1.5 step up), set 180 deg apart to preload the mesh
Number of actuators	2
Expected actuator force / torque	Carrier torque 0.1287 Nm peak at 3 per rev, 0.0965 Nm per servo against half of stall, margin 1.982. Peak pitch link force 144.19 N
Pitch-control range	Amplitude fixed at +/-40 deg by L2. Phase commandable over 120 deg (+/-60)



6.2 Pitch-Angle Relationship

Rotor azimuth definition:

Azimuth ψ is measured from the offset link direction, rotor turning counter-clockwise viewed from the non-drive end. Module vertical sits 9.08 deg further round, which trims out the cycle mean side force at zero command.

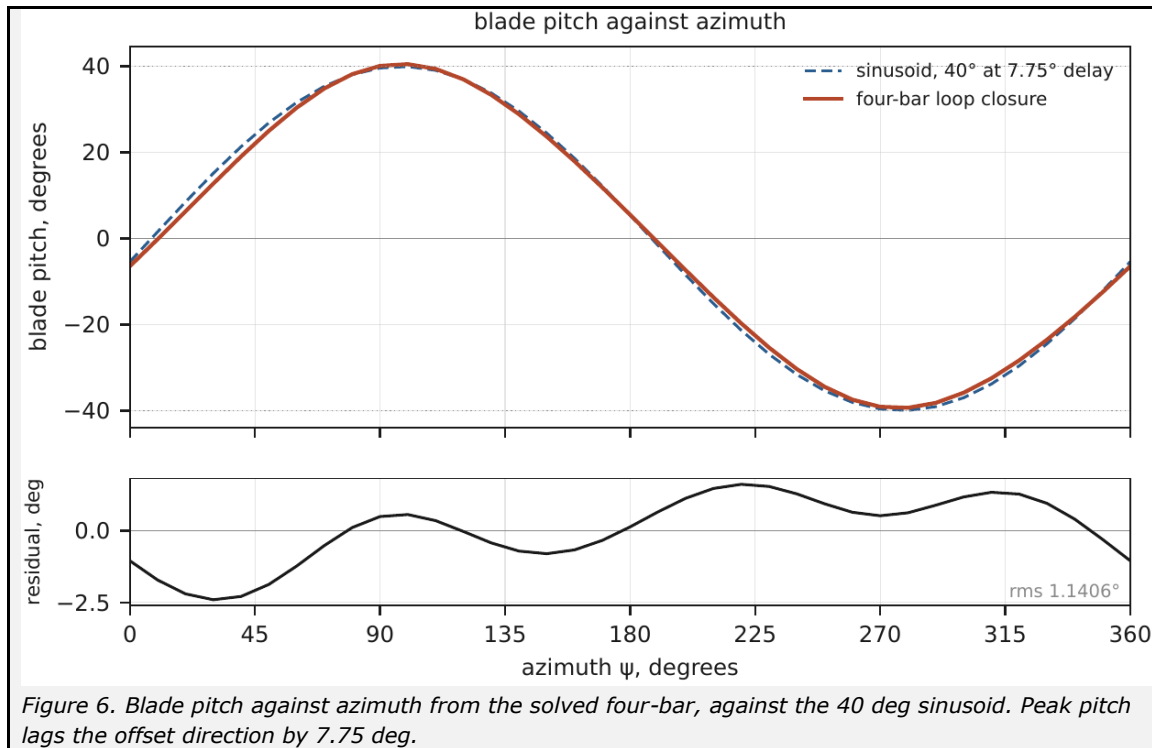
Pitch-angle relationship:

Solved from loop closure rather than prescribed: $R u(\psi) + a u(\alpha) + l u(\gamma) = e u(\phi)$, and pitch $\theta = \alpha - \psi - \theta_0$.

It stays close to $\theta = 40 \cos(\psi - 90 - \phi_c - 7.75) \text{ deg}$, with an rms deviation of 1.1406 deg. That deviation is what the load model runs on and is where part of the side force comes from.

Definition of variables:

$R = 110.0 \text{ mm}$ rotor arm; $a = 18.0 \text{ mm}$ horn; $l = 108.0 \text{ mm}$ pitch link; $e = 11.53 \text{ mm}$ offset; $u(x)$ the unit vector at angle x ; ψ blade azimuth; α horn direction and γ pitch link direction in the fixed frame; ϕ offset direction; ϕ_c the vector command; $\theta_0 = -102.50 \text{ deg}$, the fixed horn to chord angle.



6.3 Kinematic Parameters

Parameter	Value	Unit
Degrees of freedom	1 per blade, driven by rotor rotation; 1 control input, the offset direction	--
Maximum pitch angle	+40.52	deg
Minimum pitch angle	-39.35	deg
Pitch amplitude	40 (79.9 peak to	deg

	peak)	
Phase angle	7.75 linkage lag; command range +/-60	deg
Maximum actuator displacement	41.9 (80 deg servo rotation at the 60 mm sector gear)	mm
Estimated actuator force / torque	0.0965 Nm per servo (0.1287 Nm at the carrier)	N or N m

6.4 Thrust-Vectoring Assessment

Description of how cyclic blade pitch produces thrust vectoring:

Each blade's incidence follows the geometry of its four-bar, so blades on one side of the rotor run at positive pitch and blades on the other side at negative, and the cycle averaged force points roughly along the offset direction. Turning the offset pivot about the rotor axis turns the whole pitch schedule with it, and the resultant turns by the same angle.

Expected maximum vectoring angle:

+/-60 deg from the trimmed zero, 120 deg of authority. Measured side force on similar rotors runs 10 to 35 deg [3], [7], far above our model's 9.078 deg. Indexing the carrier zero at the middle of that band, we expect roughly four fifths of the authority to stay usable.

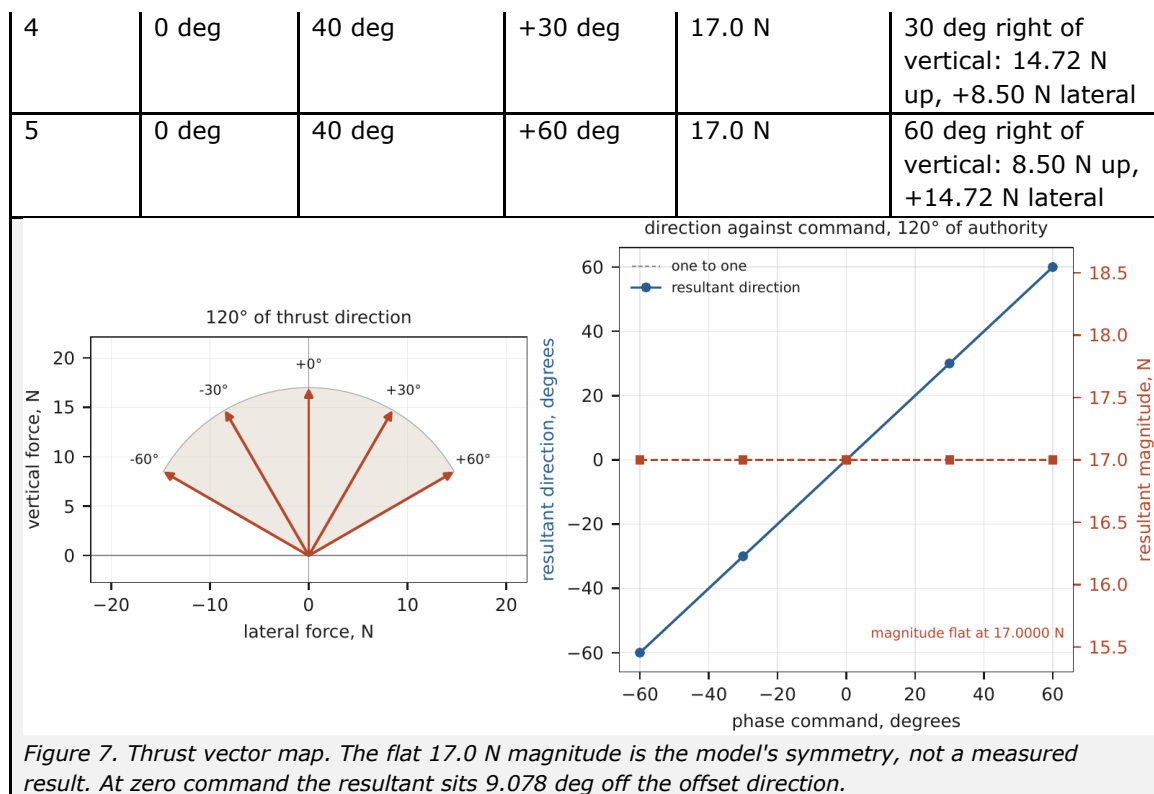
Control parameter(s):

Phase command (offset direction) sets the thrust direction. Rotor speed through the ESC sets the magnitude. Amplitude is fixed by geometry.

Expected relationship between control input and thrust direction:

One to one in the model: direction equals command and the magnitude stays 17.0 N at every command. That comes from the model's rotational symmetry and is not a measurement. The 9.078 deg tilt at zero command (7.75 deg linkage lag, about 1.33 deg aerodynamic) is a bias removed by indexing, and because the tilt moves with rpm, a magnitude change needs a small phase correction.

Case	Mean Pitch	Pitch Amplitude	Phase Angle	Estimated Thrust	Expected Thrust Direction
1	0 deg	40 deg	-60 deg	17.0 N	60 deg left of vertical: 8.50 N up, -14.72 N lateral
2	0 deg	40 deg	-30 deg	17.0 N	30 deg left of vertical: 14.72 N up, -8.50 N lateral
3	0 deg	40 deg	+0 deg	17.0 N	vertical: 17.00 N up, +0.00 N lateral



7. Powertrain Selection

7.1 Motor and Actuator Selection

Item	Details
Motor manufacturer / model	T-Motor Antigravity MN5006 KV450
KV rating	450 rpm/V
Operating voltage	8S pack: 29.6 V nominal, 28.4427 V loaded, 33.6 V charged. The windings see 23.2293 V at the design point, inside the 4-6S catalogue window of 25.2 V
Rated / maximum current	26.0 A for 180 s; 20.8 A continuous after the 0.80 derate; 19.2876 A at the design point
Maximum power	650 W for 180 s; 520 W derated continuous; 483.2 W at the design point
Motor mass	106 g with cable
Selection basis	The only one of 5 screened motors that holds power, torque (0.3902 of 0.4414 Nm), current and speed (9932 of 12799 rpm) at once, all on continuous ratings

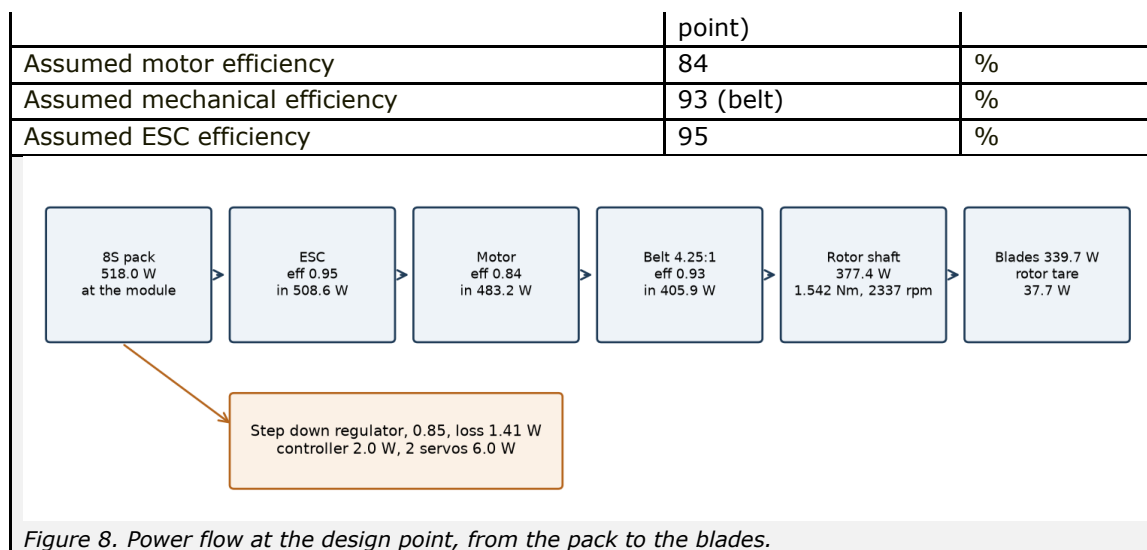
Pitch actuator model	Digital metal gear servo, 20 g class, from a supplier listing. The exact part is fixed against a datasheet in Stage 2
Pitch actuator rating	3.9 kgf cm (0.3825 Nm) stall at 6 V; slew 0.1733 s over the range
Pitch actuator mass	20.0 g each, 40.0 g for 2

7.2 Powertrain Components

Component	Proposed Specification / Model	Mass	Rating	Selection Basis
Motor	T-Motor MN5006 KV450	106.0 g	650 W, 26 A for 180 s	Datasheet; holds the design point continuously
ESC	40 A 8S class brushless ESC	19.5 g	40 A, 8S	Above the 19.29 A design current and the 33.6 V charged pack
Pitch actuator	2 x 20 g digital metal gear servo	40.0 g	0.3825 Nm stall	Margin 1.982 on the carrier ripple
Bearings	2 x 61802 main, 12 x 693ZZ pitch, 2 x MR128ZZ carrier	36.0 g	693ZZ C0 216.985 N by ISO 76	Pitch bearings duplexed, 4 per blade, for the centrifugal pull
Shaft	CFRP tube 16 x 1.5 mm, 340 mm, bonded 7075-T6 plugs	59.2 g	24.9563 Nm torsion	Torsion margin 16.18
Controller / electronics	Matek F411-WSE class board, plus a step down regulator 42 V in, 12 V out	18.5 g	Board 6-30 V input	The 8S pack exceeds the board's 30 V, so the regulator is required
Other	HTD-3M belt, 16 and 68 tooth pulleys, 4.25:1, tensioner	60.2 g	efficiency 0.93 assumed	The rotor needs 1.542 Nm; the motor gives 0.3902

7.3 Power Requirement

Parameter	Value	Unit
Estimated aerodynamic power	339.7	W
Estimated shaft power	377.4	W
Estimated electrical power	518.0 (module); 508.6 at the ESC	W
Estimated maximum current	19.29 (motor, design)	A



8. Mass Budget and Thrust-to-Weight Assessment

8.1 Mass Budget

Component	Quantity	Unit Mass	Total Mass	Basis
Blades	3	31.75 g	95.24 g	PMI foam core at 52 kg/m ³ and 88% fill, 2 plies of 60 gsm carbon twill, an 8.71 mm CFRP spar on the pitch axis and two 7075-T6 root fittings, integrated over the NACA 0020 section
Blade hubs	2 hub sets	31.32 g	62.65 g	6 CFRP spider arms of 14 x 1.5 mm, 2 split 7075-T6 bosses of 22 mm OD, 6 7075-T6 clevis brackets
Shaft	1	-	59.21 g	Roll wrapped CFRP tube, 16 mm OD, 1.5 mm wall, 340 mm, with bonded 7075-T6 end plugs forming the 15 mm journals

Bearings	16	-	36.00 g	2 x 61802 main (8.0 g each), 12 x 693ZZ pitch (1.3 g each), 2 x MR128ZZ carrier (2.2 g each), supplier listings
Frame	1	-	100.58 g	2 machined 7075-T6 bearing blocks, 4 CFRP tubes of 8 mm OD, a 2.5 mm 7075-T6 motor plate, and a visible 15.0 g reserve for gussets, clamps and trays not yet drawn
Pitch mechanism	1 set	-	36.12 g	3 CFRP pitch links with rod ends, 3 7075-T6 horns, the offset post, a 6061-T6 carrier ring with its 40 mm gear and the 60 mm sector gear
Actuator	2	20.00 g	40.00 g	20 g class digital metal gear servos, supplier listing
Motor	1	106.00 g	106.00 g	T-Motor MN5006 KV450 datasheet, with cable
ESC	1	19.50 g	19.50 g	40 A 8S class ESC with leads, supplier listing
Fasteners	about 40	-	21.50 g	M3 screws, washers and threaded inserts, plus epoxy paste at the bonded joints. Both are allowances
Mounting hardware	4	1.80 g	7.20 g	7075-T6 lugs at the 4 airframe mount points
Transmission	1 set	-	60.21 g	HTD-3M 16 and 68 tooth 6061-T6 pulleys, 9 mm by 375 mm belt, idler tensioner. 4.25 to 1
Other	3	-	43.70 g	8.5 g controller board, 10.0 g step down regulator allowance, 25.2 g wiring harness allowance

Total	34 lines	-	687.91 g	Conservative column 775.74 g (+12.8%)
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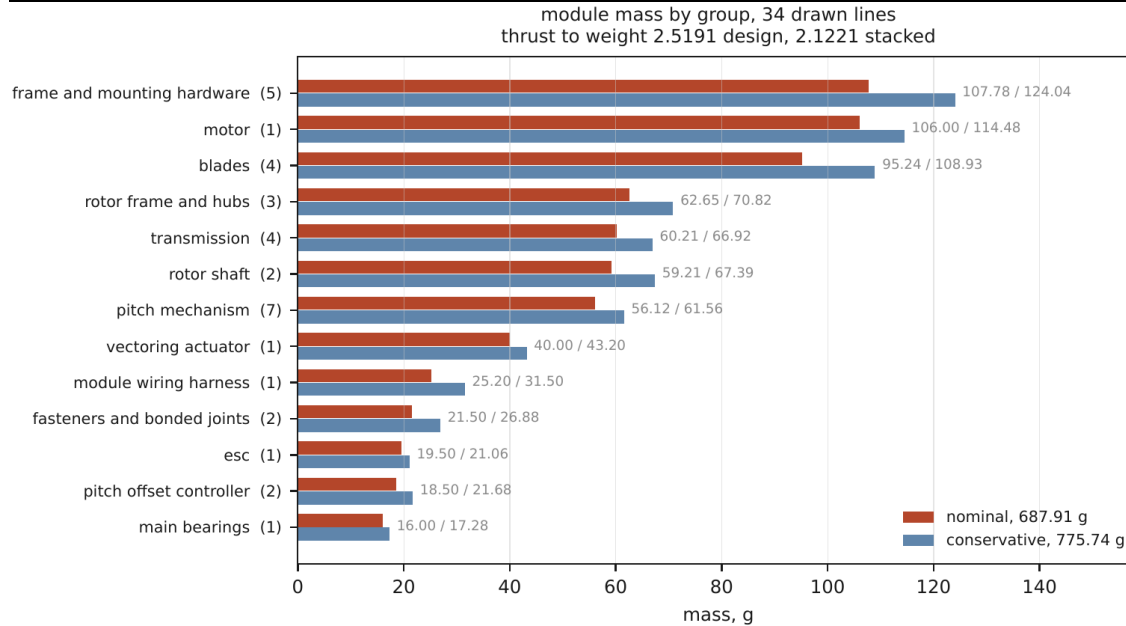


Figure 9. Module mass by group, nominal and conservative. The full 34 line budget is in the supporting files.

8.2 Thrust-to-Weight Estimate

Parameter	Value	Unit
Estimated maximum thrust	17.0 (design)	N
Estimated module mass	0.68791	kg
Estimated module weight	6.7484	N
Estimated T/W ratio	2.5191	--
Estimated uncertainty / confidence	2.12 to 2.52 across the four cases below. Medium confidence	--

Basis of thrust estimate:

Blade area coefficient 0.6055 held below Kellen's measured 0.6648 on this shape family, at 2337 rpm (section 5.3). The low case, 16.1493 N, takes 5 percent off for blade flexibility, most of which is the calculated 2.2346 deg torsional wind up.

Basis of mass estimate:

34 budget lines, each a drawn section, a catalogue part or a stated allowance, with growth by line class. It is not yet CAD mass properties or weighed parts.

Four cases are reported because they tell different stories. Design point: 17.0 N on 687.91 g gives 2.5191. Low coefficient alone: 2.3931. Conservative mass alone: 2.2339. Both together: 2.1221.

This is a preliminary pass with an open compliance risk. The design point clears 2.5 by 0.8 percent and the mass estimate isn't accurate to 0.8 percent. The conservative column would have to lose 117.26 g for the worst case to clear 2.5, so Stage 2 treats 658.48 g on that column as a gate and replaces the estimate with CAD mass properties, weighed parts and a thrust test.

9. Preliminary Structural Assessment

A detailed FEA is not required at Stage 1. The objective is to establish that the proposed structure is physically plausible and that the principal load-bearing components have been considered.

9.1 Structural Configuration

Structural arrangement:

Three sandwich blades pinned at both ends through root fittings and pitch bearings to clevis brackets on six CFRP spider arms, clamped to the CFRP shaft by two 7075-T6 split bosses. The shaft runs in two 61802 bearings in 7075-T6 blocks on four CFRP frame tubes, with the motor on a 7075-T6 plate below and four lugs to the airframe.

Critical load-bearing components:

The blade in combined centrifugal and aerodynamic bending; the root fittings and pitch bearings, which carry 209.161 N of centrifugal pull per blade; the pitch horn at 144.19 N of link force; the rotor shaft at 1.542 Nm plus a 66.502 N belt side load.

Principal load paths:

Centrifugal: blade, root fittings, pitch bearings, root brackets, spider arms, hub boss.

Thrust and reaction torque: hub boss, shaft, main bearings, bearing blocks, frame tubes, 4 mount lugs.

Pitch control: horn, pitch link, offset post, carrier bearings, frame.

9.2 Preliminary Strength Assessment

Component	Material	Critical Load	Preliminary Stress	Allowable Stress	FOS
Blade	PMI foam core, 2 ply carbon twill skin, CFRP spar	8.4153 Nm combined bending (12.1181 at 1.20 overspeed)	71.7 MPa, skin	215.3 MPa, skin wrinkling	3.00 (2.08 at overspeed)
Shaft	CFRP tube	1.542 Nm	5.56 MPa	55 MPa shear	9.90

	16 x 1.5 mm	torque + 1.995 Nm bending	combined shear		
Frame	CFRP tubes, 7075-T6 blocks and lugs	worst lug about 13.4 N	not computed	5.8 kN lug net section	over 100; stiffness governs, not strength
Hub	7075-T6 boss, 693ZZ root bearings	209.161 N per blade (301.192 at overspeed)	load check on bearing rating	781.148 N (4 bearings, ISO 76)	3.73 (2.59 at overspeed)
Pitch linkage	7075-T6 horn, CFRP link	144.19 N	121.7 MPa, horn	400 MPa (7075-T6)	3.29

Calculation methodology:

Closed form, in tools/structure.py. Beam bending on the blade section integrated from its own ordinates (EI 51.115 Nm², GJ 7.8058 Nm²), with the skin wrinkling stress $0.5 \times (E_{\text{skin}} E_{\text{core}} G_{\text{core}})^{1/3}$. Thin wall torsion and combined shear on the shaft, net section bending on the horn, and the ISO 76 static rating for the bearings. FOS is allowable over demand. Stresses in the table are the demand scaled onto the governing allowable.

Principal assumptions:

Blade load factor 4.0 on the mean aerodynamic force; a 1.20 overspeed case; published class allowables with no coupon; a floor of 1.5 on every margin.

Critical loading condition:

1.20 overspeed, about 2804 rpm, with the 4.0 aerodynamic peak on top: blade combined bending 12.1181 Nm, FOS 2.08, the lowest in the module.

Preliminary conclusion:

All eight computed margins clear 1.5 and the two lowest are both overspeed cases. Centrifugal pull, 209.161 N per blade, is 9.2 times the mean aerodynamic force, so this rotor is a centrifugal machine first. Not yet analysed: the bonded root joint itself, fatigue, the shaft's critical speed and stress concentrations. Those are Stage 2 FEA and coupon work.

10. Materials, Manufacturing, and Assembly

10.1 Material Selection

Component	Proposed Material	Key Properties	Reason for Selection
Blade	PMI foam (Rohacell 51 IG class) core, 2 plies of 60 gsm carbon twill	Foam 52 kg/m ³ , E 70 MPa, G 19 MPa; skin E 60 GPa	Skin wrinkling over the core sets the blade allowable, so the foam is structural; light enough to keep centrifugal load down
Frame	CFRP tube, 8 mm OD; 7075-T6 blocks and lugs	CFRP E 130 GPa, 1550 kg/m ³ ; 7075-T6 400 MPa, 2810 kg/m ³	Stiffness per gram for the tubes; machinable bores for the blocks
Shaft	Roll wrapped CFRP tube with bonded 7075-T6 plugs	Shear allowable 55 MPa, 1550 kg/m ³	Torsion margin 16.18 at 36 g; metal only at the journals
Hub	7075-T6 split boss, CFRP spider arms	400 MPa, 2810 kg/m ³	Clamps on the tube without crushing it; carries the root brackets
Pitch mechanism	6061-T6 carrier ring and sector gear; 693ZZ and MR128ZZ bearings	240 MPa, 2700 kg/m ³	Stiffness parts, easy to machine and gear cut
Linkages	CFRP tube 4 mm OD pitch links with rod ends; 7075-T6 horns	7075-T6 400 MPa	The horn governs the link path at FOS 3.29

10.2 Manufacturing and Assembly

Component	Proposed Manufacturing Method	Key Considerations
Blade	CNC profiled foam halves with the spar channel cut; 2 plies wet laid in a machined two part mould, vacuum bagged, room temperature cure; 7075-T6 root fittings bonded and pinned	Mould quality and blade mass matched within 0.5 g across the set
Frame	CFRP tubes cut to length; bearing blocks CNC milled with both bores in one setup; bolted	Bearing bore alignment, 0.010 mm
Shaft	Plugs bonded into the tube first, then both journals ground in one setup	Journal run out under 0.02 mm; grinding after bonding keeps them concentric
Hub	Turned and milled 7075-T6 boss, reamed bore; spider	Bore to shaft 0.015 mm; clamp without crushing the tube

	arms CNC routed from 1.5 mm CFRP plate	
Pitch mechanism	Turned 6061-T6 carrier and sector gear, gear teeth cut as job work; CNC 7075-T6 horns and post; rod ends bought	Backlash 0.05 mm; pitch checked against the solved schedule at 12 azimuths

Expected manufacturing constraints:

The PMI foam is a 4 week import with no Indian stockist, and it gates the build, so it is ordered first. The gear pair is the only job needing a cutter a local shop won't keep, and at a quantity of one it is priced by setup. Two joints, the shaft plugs and the blade root, can't be reworked once cured.

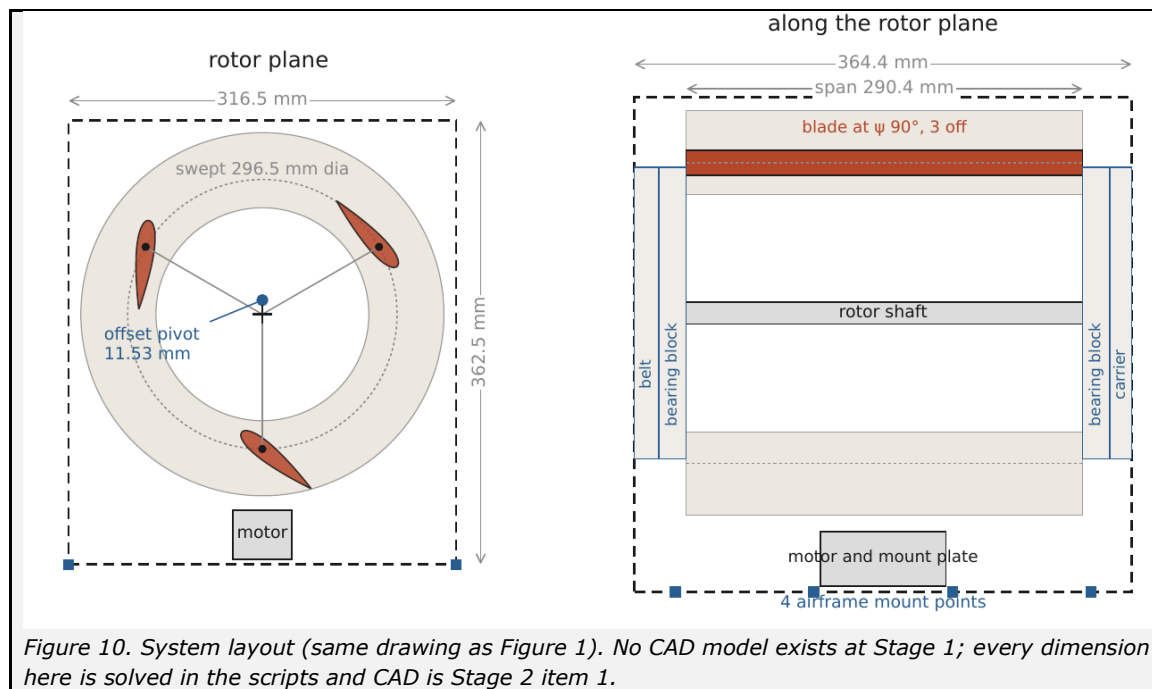
Availability of materials/components:

Motor, servos, bearings, belt, ESC and CFRP tube are stocked by Indian distributors. The bill of materials totals 68,830 INR (40,030 bought parts and material, 28,800 tooling and fabrication), at distributor list prices. Nothing has been quoted yet.

Preliminary assembly approach:

Bond the shaft plugs and grind the journals; build and weigh the 3 blades; mate the spiders to the shaft and hang the blades on their pitch bearings; fit the main bearings and blocks to the frame and drop the rotor in; fit the carrier, offset post and pitch links, setting link lengths against the schedule; then motor, pulleys and belt, tensioned last; then ESC, controller, servos and harness. Before any spin: blade masses matched, run out measured, pitch checked at 12 azimuths, a static pull on one attachment above 301 N, and a first spin in four steps with current logged.

11. CAD and System Packaging



11.1 Major Dimensions

Parameter	Dimension	Unit
Overall length	364.4 (along the rotor axis)	mm
Overall width	316.5	mm
Overall height	362.5 (includes the 46 mm motor stack)	mm
Rotor diameter	220.0 (296.5 swept)	mm
Rotor width	290.4	mm
Mounting dimensions	4 lugs on a 320 x 240 pattern	mm
Motor envelope	46 axial stack below the rotor, on the drive end plate	mm
Actuator envelope	Non-drive end: 16 axial phasing carrier bay; 2 servo cases on the frame, not yet dimensioned	mm

11.2 System Integration

Item	Description
------	-------------

Motor arrangement	One outrunner under the rotor at the drive end, belt up to a 68 tooth rotor pulley
Bearing arrangement	2 x 61802 on the shaft end plugs in 7075-T6 blocks; 12 x 693ZZ at the blade roots; 2 x MR128ZZ carrying the carrier
Pitch mechanism	All at the non-drive end: offset post from the phasing carrier, 3 pitch links to the horns
Actuator arrangement	2 servos 180 deg apart on sector gears, outside everything that rotates
Mounting interface	4 lugs in the plane of the rotor axis. 4 rather than 3 because the airframe sees 1.542 Nm of steady reaction torque
Potential interference	The pitch links pass 0.019 mm from the rotor axis, so nothing coaxial may sit in that plane. Blade sweep annulus 86.92 to 148.25 mm kept clear. Neighbour clearance 98.47 mm
Assembly considerations	Bonded joints first, then rotor into frame, mechanism, drive and electronics last
Maintenance/accessibility	Carrier and servos come off the non-drive end without disturbing the rotor; belt reachable at the drive end; each blade comes out once its link is unpinned
Critical dimensions/tolerances	Journal run out 0.02 mm; bearing bores 0.010 mm; pitch link free length 0.05 mm; gear backlash 0.05 mm

12. Design Risks and Open Technical Questions

12.1 Risk Assessment

Risk / Failure Mode	Cause	Consequence	Probability	Severity	Mitigation
Thrust below 17.0 N	Transferred coefficient wrong for this rotor; high inflow ratio 0.3871	T/W under 2.5; 10 N still met at 16.15 N	Medium	High	Transient CFD, then a load cell run
Mass growth	Wet layup blades come out	T/W under 2.5 (margin is 0.8%)	High	High	CAD mass properties, weighed parts,

	heavy; undrawn frame parts use up the 15 g reserve				658.48 g gate
Motor overheats in continuous running	0.80 derate has no source; motor at 93% of it	Lower thrust or a heavier motor	Medium	Medium	Dynamometer run at working current, early in Stage 2
Pitch bearing wear (false brinelling)	80 deg oscillation, balls never recirculate (ratio 0.5533)	Pitch play, loss of schedule	Medium	Medium	Run to failure at speed on the flight grease
Side force larger than modelled	Quasi steady model; measured 10 to 35 deg on similar rotors	About a fifth of vectoring authority lost	High	Low	Index carrier zero; load cell calibration across speed
Blade root joint failure	301.192 N through a bonded joint not yet qualified	Blade release	Low	High	Bond shear coupons, static pull above 301 N before any spin, spin behind a guard
Foam supply delay	4 week import, no Indian source	Build slips	Medium	Medium	Order first; fallback is a thicker skin and a rerun

12.2 Open Technical Questions

List key technical questions that require further detailed analysis, simulation, or validation.

1. Does the transferred coefficient hold on this rotor at inflow ratio 0.3871? Transient CFD and a load cell run.
2. How large is the real side force, and how does it move with rpm? It sets usable vectoring authority.
3. What life do the 693ZZ pitch bearings have in 80 deg oscillation at 39 Hz?

4. How strong is the bonded blade root in shear, peel and 3 per rev fatigue?
5. Where is the shaft's first bending mode against 2337 rpm, and does the MN5006 hold its rating at the 8S duty?

13. Design Development and Engineering Decisions

13.1 Design Evolution

Version	Date	Major Change	Reason / Evidence	Outcome
V1	29-31 Aug 2026	Geometry frozen: R 110 mm, 3 blades, 18 N design, 6S pack, 3.5:1 belt	Kellen's shape family and the week 2 radius sweep	580.05 g, T/W 3.1633 design, 2.2525 stacked
V2	1 Sep 2026	Figure of merit corrected 0.6 to 0.5215; design 17 N; pack 8S; belt 4.25:1; 20 g servos; four-bar 18 x 108 mm; pitch bearings doubled; six mass lines redrawn	FM must scale with the lowered coefficient; at 6S the motor capped at 392 W against 406 W needed	677.91 g, T/W 2.5563 design, 2.1569 stacked
V3	4 Sep 2026	10 g step down regulator added for the pitch controller	The controller is rated 6 to 30 V and a charged 8S pack reaches 33.6 V	687.91 g, T/W 2.5191 design, 2.1221 stacked

13.2 Significant Design Decisions

Item	Decision 1
Design question	Design thrust and pack voltage after the power correction
Options considered	Keep 18 N on a bigger motor; drop thrust on the same motor; raise the pack voltage
Selected option	17 N on the MN5006, 8S pack, 4.25:1 belt
Basis for selection	Output power a motor can give is capped at the speed rule x loaded pack voltage x continuous current, and KV cancels, so gearing can't move it. 8S lifts the cap; 17 N is where every drive line keeps 7% or better
Supporting calculation /	At 8S the cap is 532.4 W against 405.9 W wanted. Motor

evidence	at 483.2 of 520 W, 19.29 of 20.8 A, 0.3902 of 0.4414 Nm, 9932 of 12799 rpm
Item	Decision 2
Design question	Should the blade be chordwise balanced?
Options considered	Balance with nose ballast at the pitch axis; leave it unbalanced
Selected option	Leave it unbalanced
Basis for selection	Balancing lifts the pitch link margin from 3.29 to 6.20 but costs mass on a case that already misses
Supporting calculation / evidence	Balancing takes the stacked T/W from 2.1221 to 2.0292; the unbalanced link path still clears at 3.29

13.3 Current Design Limitations

1. Thrust and the side force rest on a transferred coefficient and a quasi steady model. Nothing has been measured on this rotor.
2. Mass is a line by line estimate, not CAD mass properties or weighed parts, and the pass at the design point is 0.8 percent.
3. No CAD, FEA or CFD exists yet. Structure is closed form and the root joint, fatigue and rotor dynamics are not analysed.
4. The efficiency chain and the motor derate are assumed, and the motor figure is the sensitive one.
5. The team is three programmers. Fabrication and testing experience has to come from new members, a mentor or the workshop.

14. References and Sources

All externally sourced numerical values, component specifications, material properties, aerodynamic data, and design correlations should be traceable to their respective sources.

No.	Reference / Source	Link / DOI / Datasheet	Application in Design
1	Kellen, A. J. Performance Measurements on a UAV-Scale Cycloidal Rotor in Hover. MS thesis, Texas A&M University, 2019	https://hdl.handle.net/1969.1/184958	Shape family; coefficient 0.6648; FM 0.6; solidity and Reynolds bands; horn length
2	Benedict, M. Fundamental Understanding of the Cycloidal-Rotor Concept for Micro Air Vehicle Applications. PhD	https://hdl.handle.net/1903/11257	Coefficients 0.7211 and 0.8114; power loading 0.062 N/W; 10% tare; blade count

	dissertation, University of Maryland, 2010		
3	Sirohi, J., Parsons, E. and Chopra, I. Hover performance of a cycloidal rotor for a micro air vehicle. J. American Helicopter Society, 2007	journal article	Side force angle, about 10 deg
4	Xisto, C., Leger, J., Pascoa, J. et al. Parametric analysis of a large-scale cycloidal rotor in hovering conditions. J. Aerospace Engineering, 2016	journal article	Parametric context for blade number and pitch
5	Runco, C. and Benedict, M. Design, development, and flight testing of a 70-gram micro quad-cyclocopter. Int. J. Micro Air Vehicles 15, 2023	open access	Module T/W benchmark; centrifugal to aerodynamic load ratio
6	Shrestha, E., Benedict, M. et al. Understanding upward scalability of cycloidal rotors for large-scale UAS applications. JAHS 67(4), 2022	journal article (summary read)	Reynolds invariance; blade mass scaling
7	Adams, Z., Benedict, M., Hrishikeshavan, V. and Chopra, I. Design, development, and flight test of a small-scale cyclogyro UAV utilizing a novel cam-based passive blade pitching mechanism. Int. J. Micro Air Vehicles 5(2), 2013	journal article (summary read)	Side force 15 to 35 deg
8	Alsabri, A. A. M. et al. Independent effects of blade number and solidity on cyclorotor hover performance. Aerospace 13(9), 765, 2026	https://doi.org/10.3390/aerospace13090765 (summary read)	Peak to mean blade load 3 to 4
9	T-Motor Antigravity MN5006 KV450 datasheet	manufacturer datasheet (PDF)	650 W / 26 A for 180 s, 106 g, 60 mOhm, 4-6S
10	ISO 76, Rolling bearings: static load ratings	standard	693ZZ static rating, 216.985 N

15. Digital / AI Tools and Design Integrity

15.1 Digital / AI Tools Used

Tool	Purpose	Nature / Extent of Use
Claude (Anthropic), used through Claude Code	Writing and checking the Python calculation scripts, drafting report text, reading and summarising sources, internal design reviews	Extensive. Most of the calculation code and text was produced with it under the team's direction. Every number is recomputed by the scripts from one data file, and the team takes responsibility for all of it
OpenAI Codex	Independent audits of the report against the problem statement	Several review rounds. Each finding was checked, then fixed or answered in the design record
Python 3, NumPy, Matplotlib	All calculations and figures	Open source libraries
Pandoc, XeLaTeX, Microsoft Word	Document preparation	Formatting only

15.2 Design Integrity Declaration

We confirm that the engineering design, calculations, CAD development, analysis methodology, interpretation of results, and technical decisions presented in this report have been reviewed and verified by the team.

The team accepts responsibility for the technical accuracy, validity, and integrity of the submitted work.

The team understands that the evaluation committee may request supporting calculation files, CAD models, design-history records, or other technical evidence and may require team members to explain, reproduce, modify, or defend the technical work presented in this report.

Team Leader:	Kartik Shirode
Signature:	
Date:	27 September 2026

16. Final Stage 1 Design Summary

Parameter	Final Stage 1 Value	Unit
Rotor diameter	220.0	mm
Rotor width	290.4	mm
Number of blades	3	--
Blade chord	72.6	mm
Operating RPM	2337	rpm
Maximum blade pitch	40.52	deg
Pitch amplitude	40	deg
Maximum vectoring angle	+/-60	deg
Estimated thrust	17.0	N
Estimated torque	1.542	N m
Estimated shaft power	377.4	W
Estimated electrical power	518.0	W
Estimated module mass	0.688	kg
Estimated T/W ratio	2.5191	--
Primary blade material	PMI foam core, carbon twill skin, CFRP spar	--
Primary frame material	CFRP tube and 7075-T6	--
Motor	T-Motor MN5006 KV450	--
Pitch actuator	2 x 20 g digital metal gear servo	--

Why is the proposed design expected to meet the 10 N thrust requirement?

Design thrust is 17.0 N on a coefficient held below the value measured on this shape family, at a Reynolds number inside the measured band. Even with 5 percent off for blade flexibility it is 16.15 N, 61 percent above the requirement. The motor holds the design point on continuous ratings.

Why is the proposed design expected to achieve $T/W > 2.5$?

At the design point it does, at 2.5191, but only by 0.8 percent, so we don't claim it with confidence. The three downside cases fall to 2.39, 2.23 and 2.12. We publish the 117.26 g that would close the worst case, and Stage 2 replaces the estimate with measured mass and thrust before anything else. The routes to more margin are a measured coefficient and a lower KV motor on more cells.

What are the three most important assumptions underlying the current design?

1. The blade area coefficient 0.6055 transfers to this rotor.
2. The mass budget holds: 34 lines with growth by class, nothing yet weighed.
3. The efficiency chain (0.93, 0.84, 0.95) and the 0.80 motor derate.

What are the most important technical questions still to be resolved?

1. Measured thrust and side force on this rotor.
2. Measured module mass, starting from CAD mass properties.

3. Strength and life of the bonded blade root and the oscillating pitch bearings.

Overall assessment of Stage 1 design readiness:

Preliminary design closed on paper. Geometry, kinematics, loads, mass, power and cost are consistent with each other and recomputed from one file. It is not build ready: thrust to weight is an open compliance risk, and CAD, CFD, FEA and testing are the first Stage 2 work.

Appendix A - Supporting Calculations

Attach detailed hand calculations, spreadsheet outputs, MATLAB/Python calculations, or other supporting engineering calculations.

Supporting files in the zip, folder 03_Calculations: stage-1/design/numbers.json holds every input and result; tools/linkage.py solves the four-bar, the pitch schedule, the aerodynamic loads and the vector map; tools/structure.py builds the blade section, the margins and the mass budget. Both are plain Python 3 with no third party packages. Running `python tools/linkage.py --write` and then `python tools/structure.py --write` regenerates numbers.json byte for byte. The 32 page detailed design note in folder 02_Detailed_Design_Note carries every calculation in prose.

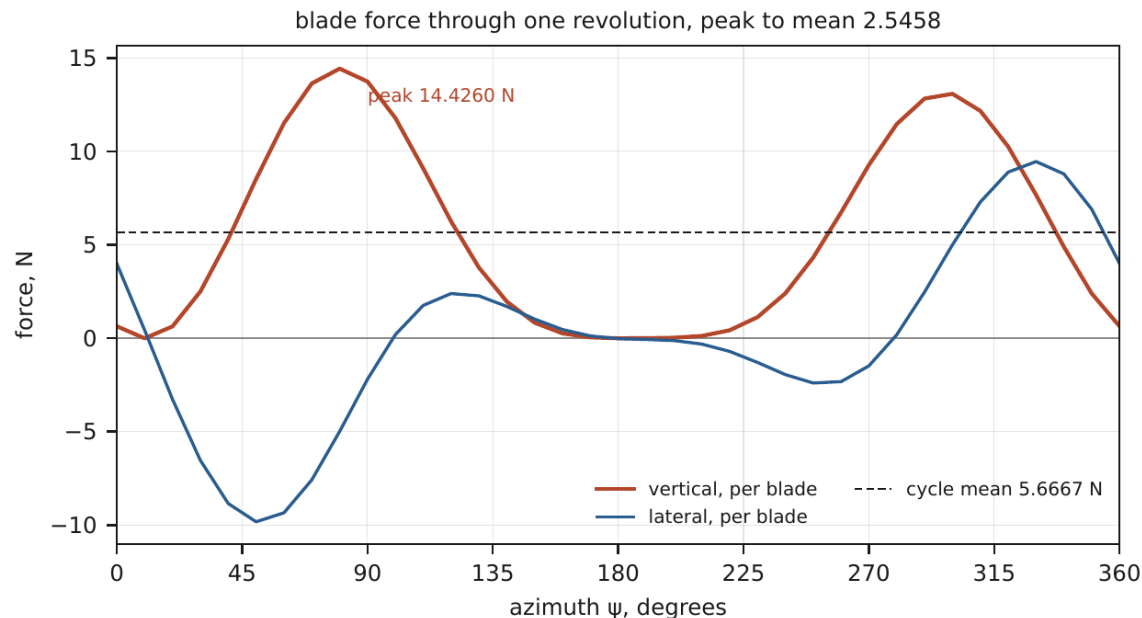


Figure A1. Blade force against azimuth on the solved pitch schedule. Peak 14.4260 N against a mean of 5.6667 N.

Appendix B - Preliminary CAD

Attach additional CAD views, drawings, dimensions, and component layouts.

No CAD model exists at Stage 1. The dimensioned layout is Figure 1, the blade section Figure 3

and the linkage Figure 5, all drawn from the same numbers file. The packaging and interface definition is in 04_Design_Record/09-packaging-and-integration.md.

Appendix C - Design Development Evidence

Attach preliminary sketches, earlier design versions, calculation iterations, design-review records, photographs, or other evidence of design development.

Folder 04_Design_Record holds the design documents for each item, the evidence ledger that grades every borrowed number, and the decision log, 72 dated entries recording each change in section 13.1 and why it was made.

Appendix D - Additional Technical References

Attach additional papers, datasheets, technical reports, standards, and other sources.

Not read and not used as evidence, listed for Stage 2: Ramsey, R. A., Development and Flight Testing of a 25-Kilogram Quad-Cyclocopter, MS thesis, Texas A&M, 2022; Heimerl, Halder, Benedict et al., Experimental and Computational Investigation of a UAV-Scale Cycloidal Rotor in Forward Flight, VFS 77th Annual Forum, which measured instantaneous blade forces.

Stage 1 Evaluation Summary

Evaluation Criterion	Weight	Score
Engineering requirements and design targets	5	
Concept development and selection	8	
Rotor and blade design	10	
Preliminary aerodynamic analysis	15	
Blade-pitch kinematics and thrust vectoring	15	
Powertrain and power requirement	8	
Mass budget and T/W assessment	12	
Preliminary structural assessment	5	
Materials, manufacturing, and assembly	5	

CAD and system packaging	5	
Risk assessment and open technical questions	4	
Design development and engineering decisions	8	
Technical clarity and completeness	5	
Total	100	

Stage 1 Evaluation Philosophy

Stage 1 focuses on engineering reasoning, preliminary calculations, feasibility, and technical credibility. Detailed CFD, detailed FEA, optimization, prototype development, and experimental validation are not expected at this stage.

The quality of a submission will be assessed primarily on:

- Soundness of engineering assumptions.
- Correctness and transparency of preliminary calculations.
- Physical plausibility of predicted performance.
- Understanding of blade-pitch kinematics and thrust vectoring.
- Credibility of the mass and power estimates.
- Consistency between geometry, operating conditions, and predicted performance.
- Quality of engineering judgement and design trade-offs.
- Evidence that the team understands the proposed design and its limitations.

A technically simple design supported by transparent and defensible engineering calculations is preferred over a complex design supported by insufficient justification.